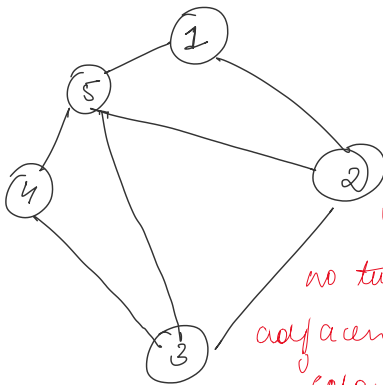


Graph colouring

Saturday, 14 February 2026 8:39 PM



$$M = 3$$

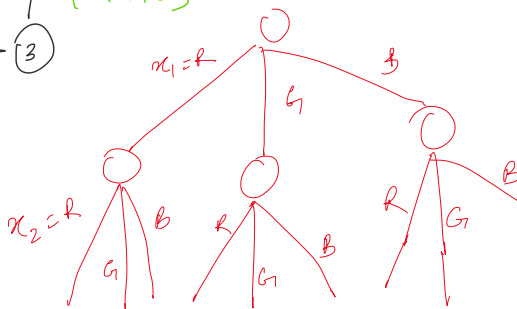
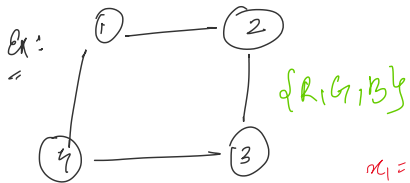
Red, Green, Blue

Q1 we have to colour the vertices of graph such that no two neighbouring vertices or adjacent vertices have same colour.

1 2 3 4 5
R G R G B *

Q1 colouring decision problem

Q2 colouring optimization problem



$x_1 = 3$

The graph coloring problem can be used in Map where we have to colour the map & no two region should have same colour.

